Bandi Sai Prasanna

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	7.01
NARAYANA JUNIOR COLLEGE, TIRUPATI	Class XII (BIEAP)	2024	98.40%
NARAYANA E.M HIGH SCHOOL, M.R.PALLI CHITTOOR DISTRICT ,	Class X (BSEAP)	2022	97.00%

PROJECTS

TCP Echo Client-Server (C, POSIX Sockets)

- Implemented client-server communication over TCP with DNS hostname resolution **(getaddrinfo)**, multi-threaded connection handling (pthreads) for concurrency, and UDP multicast for broadcast-style server monitoring.
- Validated implementation using packet capture tools (tcpdump, Wireshark) to inspect DNS resolution traffic.

XV6 Kernel Extension | C

- Developed kernel modules in C to extend the xv6 operating system with scheduling, virtual memory, and storage management, improving execution efficiency and modular system design.
- Designed and implemented a **Multi-Level Feedback Queue (MLFQ)** scheduler with four priority levels, priority boosting, and process statistics tracking to analyze scheduler performance.
- Implemented CLOCK Page Replacement, Page Fault Handling, Frame Table Management, and Swap-in/Swapout mechanism with **priority-based page eviction** for efficient dynamic memory management under varying workloads.
- Developed swapping subsystem supporting **FCFS, SSTF**, and simulated **RAID 0/1/5** storage configurations, while integrating storage with the virtual memory subsystem for efficient operation.

PostgreSQL Query Optimization & Performance Analysis | PostgreSQL.

- Optimized query execution on a 60M-row TPC-H benchmark by evaluating **B-Tree, Clustered, Composite, and Index-Only Indexes** across varying query selectivities to minimize query execution latency and improve database performance.
- Used python automation code to test the performance of query , compute selectivity, and analyze EXPLAIN (ANALYZE, FORMAT JSON) execution plans for detailed performance profiling, analysis, optimization,and analyzing processes.
- Evaluated optimizer behavior using **pg_hint_plan** to identify efficient execution strategies, compare indexing techniques, and analyze query planner decision making under varying workloads, selectivities, and execution patterns.

Object-Oriented Arbitrary Precision Arithmetic Library | OOPs

- Designed and implemented an object-oriented arbitrary precision arithmetic library supporting integers beyond native data type limits using custom data structures and algorithms from scratch independently.
- Developed **Addition, Subtraction, Multiplication, and Division** using custom numeric representations and Object-Oriented Programming (OOP) principles to ensure correctness, modularity, and extensibility.
- Validated arithmetic operations across diverse test cases, ensuring correctness, robustness, and reliable handling of **large integer computations**, including edge cases, boundary conditions, invalid inputs, and overflow scenarios.

SKILLS

Technical skills: C, C++, SQL, git, html, CSS, laTex

Soft Skills: Communication, Team Collaboration, Leadership

RELEVANT COURSES

Computer Science Courses: Data Structures and Algorithms, DBMS-1, DBMS-2, Software Development Fundamentals, Computer Architecture, Operating Systems-1, Operating Systems-2, Compilers-1

Mathematics Courses: Discrete Maths, Theory of Computation, Probability and Random Variables, Introduction to Statistics, Calculus-1, Calculus-2, Linear Algebra, Transform Techniques

POSITIONS OF RESPONSIBILITY

General Secretary, Indian Institute of Technology Hyderabad (IITH) | Current

- Represent student concerns and communicate their needs effectively to the relevant authorities.
- Understand student challenges, assist in problem-solving, and encourage innovation through collaborative initiatives at IITH.

Production Coordinator, Diesta

- Managed participant coordination, event schedules, and production-related activities.
- Supported venue arrangements and logistical operations for smooth event execution.

BTL Core Member

- Collaborated with team members in planning schedules and managing event logistics.
- Developed teamwork, communication, and organizational skills through active participation in event coordination.

EXTRACURRICULAR

Actively participated in NSS activities, including walkathons and Swachh Bharat campaigns, promoting community service and social awareness.

Elected as General Secretary in student elections by understanding student concerns, communicating effectively with peers, and representing their interests.